

Partial Result: Finite Cotype is Necessary for Nonlinear Expander Families

Automatic Math Discovery Run `fa.banach.001`

June 18, 2026

Source Problem

The source paper is Manor Mendel and Assaf Naor, *Towards a calculus for non-linear spectral gaps*, arXiv:0910.2041 [1]. On PDF page 20, the authors ask for geometric conditions on metric spaces that admit expander families with respect to those spaces. For normed spaces they write that it is tempting to conjecture that having such an expander is equivalent to having finite cotype. In the parsed source this is `data/parsed/arxiv_sources/0910.2041/zigzag-arxiv-v1.tex`, lines 1238–1247.

are natural spaces like L_∞ and L_1 for which we do not know the answer. Indeed, it may be the case that any normed space has a sufficient decay.

Another question is what are the geometric conditions on metric spaces for the existence of expanders with respect to those spaces. It is tempting to conjecture that for normed spaces, having such an expander is equivalent to having finite cotype (i.e., $\exists \varepsilon_0 > 0, n_0 \in \mathbb{N}$ such that $\ell_\infty^{n_0}$ does not embed with distortion less than $1 + \varepsilon_0$).

As was mentioned in Section [1.1](#), it is an open question whether every “classical” expander is also super-expander. In the full version we rule out the most obvious approach to prove such a result: coarsely embedding any expander family in any other expander family. We give an example of two expander families $\mathcal{F}_1$ and $\mathcal{F}_2$ such that $\mathcal{F}_1$ does not coarsely embed in $\mathcal{F}_2$.

It seems to be a challenging problem whether in [\(33\)](#) we can take $\delta = 1$. Even in the case $X = L_p$, $p \neq 2$, this seems to be unknown. A classical result of Meyer [\[24\]](#) implies that for $p \geq 2$ we can take $\delta = \frac{1}{2}$, but even this improved bound is insufficient for our purpose of constructing a “sufficiently good” base graph.

Definitions

For a finite connected graph $G = (V, E)$ and a metric space (Y, d_Y) , say that G has nonlinear Poincare constant at most Γ with respect to (Y, d_Y^2) if every map $f : V \rightarrow Y$ satisfies

$$\frac{1}{|V|^2} \sum_{u,v \in V} d_Y(f(u), f(v))^2 \leq \frac{\Gamma}{|E|} \sum_{\{u,v\} \in E} d_Y(f(u), f(v))^2.$$

A graph family is an expander family with respect to Y in this sense if the degrees are uniformly bounded, $|V| \rightarrow \infty$, and the constants Γ are uniformly bounded.

We use the Maurey–Pisier finite-representability theorem in the following standard form: a Banach space X has no finite cotype if and only if, for every n and every $\eta > 0$, ℓ_∞^n embeds linearly into X with distortion at most $1 + \eta$.

Idea of the Proof

The obstruction is already visible for $X = \ell_\infty$, as the source notes. The Maurey–Pisier theorem says that a Banach space with no finite cotype contains finite-dimensional copies of ℓ_∞^n at all scales. Every n -vertex graph embeds isometrically into ℓ_∞^n by distance coordinates. On a bounded-degree graph most vertex pairs are separated by order $\log n$, while edges remain length one. The Poincare constant must therefore grow at least quadratically in $\log n$, contradicting uniform boundedness.

The Necessary Direction

Partial Theorem. *Let X be a Banach space. Suppose there are finite connected graphs $G_k = (V_k, E_k)$ with uniformly bounded degrees, $|V_k| \rightarrow \infty$, and*

$$\sup_k \gamma(G_k, \|\cdot\|_X^2) < \infty$$

in the nonlinear Poincare sense above. Then X has finite cotype.

Proof. We prove the contrapositive. Assume that X has no finite cotype. Fix $\eta \in (0, 1)$. By Maurey–Pisier, for every n there is a linear map $T_n : \ell_\infty^n \rightarrow X$ such that

$$\|z\|_\infty \leq \|T_n z\|_X \leq (1 + \eta)\|z\|_\infty \quad (z \in \ell_\infty^n).$$

Let $G = (V, E)$ be a connected graph on n vertices with maximum degree at most D . Define the distance-coordinate embedding

$$\Phi : V \rightarrow \ell_\infty^V, \quad \Phi(v) = (d_G(v, w))_{w \in V}.$$

For $u, v \in V$, the triangle inequality gives $\|\Phi(u) - \Phi(v)\|_\infty \leq d_G(u, v)$, while taking $w = u$ gives the reverse inequality. Thus Φ is an isometric embedding of the graph metric into ℓ_∞^V . Put $f = T_n \Phi : V \rightarrow X$. Then

$$d_G(u, v) \leq \|f(u) - f(v)\|_X \leq (1 + \eta)d_G(u, v)$$

for all $u, v \in V$. In particular, every edge has X -length at most $1 + \eta$.

We also need the elementary bounded-degree distance estimate. There is a constant $c_D > 0$, depending only on D , such that for every sufficiently large connected n -vertex graph of maximum degree at most D ,

$$\frac{1}{n^2} \sum_{u, v \in V} d_G(u, v)^2 \geq c_D (\log n)^2.$$

Indeed, a ball of radius r in a graph of maximum degree D has at most $C_D(D - 1)^r$ vertices when $D \geq 3$, and at most $2r + 1$ vertices when $D \leq 2$. Taking r to be a sufficiently small constant multiple of $\log n$, at least half of the vertices are outside $B(u, r)$ for every center u , for all large n .

Therefore the Poincare constant of G with respect to X satisfies

$$\gamma(G, \|\cdot\|_X^2) \geq \frac{\frac{1}{n^2} \sum_{u, v} \|f(u) - f(v)\|_X^2}{\frac{1}{|E|} \sum_{\{u, v\} \in E} \|f(u) - f(v)\|_X^2} \geq \frac{c_D (\log n)^2}{(1 + \eta)^2}.$$

This lower bound tends to infinity with n . Hence no bounded-degree graph family with $|V_k| \rightarrow \infty$ can have uniformly bounded nonlinear Poincare constants with respect to X . This proves the contrapositive. $\square$

Scope and Limitations

The theorem proves the necessary direction of the source’s finite-cotype equivalence: existence of a normed-space expander family forces finite cotype. It does not prove the converse, namely that every Banach space of finite cotype admits such an expander family.

Novelty and Search Bounds

The run indexes were searched for arXiv:0910.2041, nonlinear spectral gaps, finite cotype, ℓ_∞ , Maurey–Pisier, and expanders with respect to Banach spaces. No prior packet for this source question or this necessary direction was found. The argument is a short standard consequence of Maurey–Pisier and the distance embedding into ℓ_∞^n , so human review should treat it as a clarification of a natural half of the conjecture, not as a solution of the full equivalence.

Human Review Recommendation

Check the normalization of the nonlinear Poincare constant against the source’s notation. The proof is invariant under the usual equivalent normalizations, but the packet states the edge-average form used in the source introduction.

References

- [1] M. Mendel and A. Naor, *Towards a calculus for non-linear spectral gaps*, arXiv:0910.2041, 2009.
- [2] B. Maurey and G. Pisier, *Series de variables aleatoires vectorielles independantes et proprietes geometriques des espaces de Banach*, Studia Math. 58 (1976), 45–90.